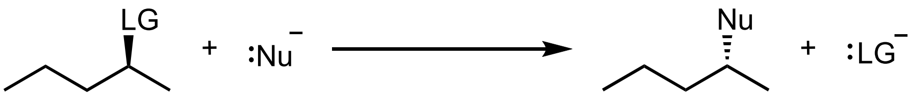
Organic Reaction Library

Chem 0310

1 Bimolecular Substitution — S_N2

6-2 through 6-11, 7-5

mechanism ✓
pg.242, 367

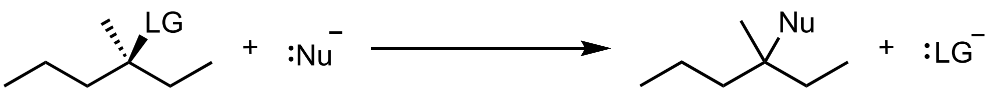


concerted, complete inversion of configuration by backside attack

2 Unimolecular Substitution — S_N1

7-1 through 7-5

mechanism ✓
pg.271, 273

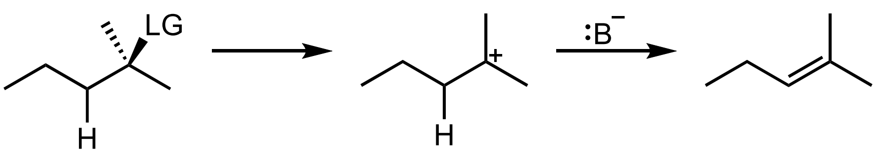


carbocation formed, stereocenter of substitution is racemized

3 Unimolecular Elimination — E1

7-6 – additional notes 9-2, 9-3, 11-7

mechanism ✓
pg.283

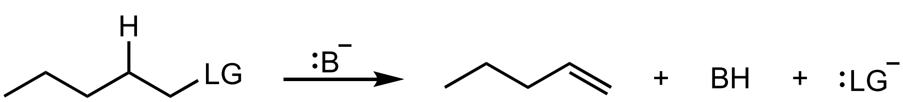


carbocation formed (rearrangement!), most substituted product(s) formed

4 Bimolecular Elimination — E2

7-7 – regio and stereochemistry 11-6

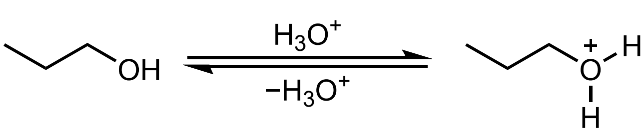
mechanism ✓
pg.285



concerted, *anti* elimination, favored by strong bases, see 70 and 71

5 Protonation of Alcohols

8-3

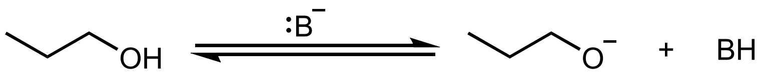


alkyloxonium ion formed

6 Deprotonation of Alcohols

mechanism ✔

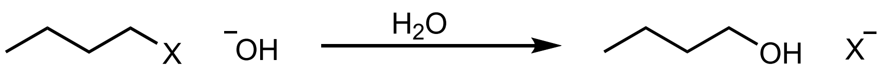
8-3, 9-1 pg.309, 310



alkoxide ion formed
Acidity of ROH: tertiary < secondary < primary < methyl

7 Hydroxide Substitution

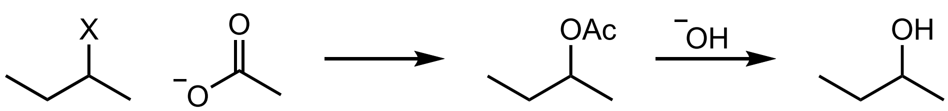
8-4



X = $Cl^- < Br^- < I^- < RSO_3^-$
primary and secondary substrates (tertiary undergoes elimination)
see S_N2 in 1

8 Ester Hydrolysis

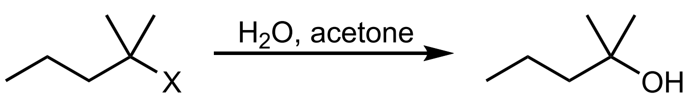
8-4 – hydrolysis in 19-9, 20-4



S_N2 with AcO^- followed by hydrolysis

9 Hydrolysis

8-4

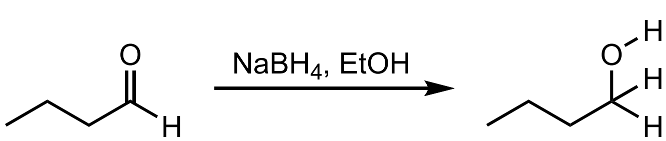


tertiary substrates undergo S_N1 2 with water

10 Borohydride Reduction of Aldehydes

mechanism ✔

8-5 pg.317

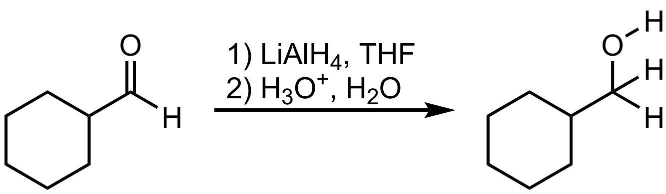


aldehydes yield primary alcohols
four equivalents of hydride furnished
Note: ternary mechanism not realistic

11 Aluminum Hydride Reduction of Aldehydes

mechanism ✓

8-5 pg.318

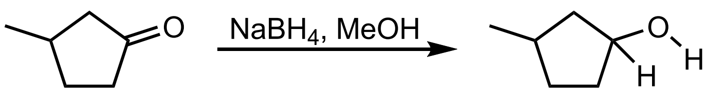


aldehydes yield primary alcohols
two-step process with acidic workup
four equivalents of hydride furnished

12 Borohydride Reduction of Ketones

mechanism ✓

8-5 pg.317

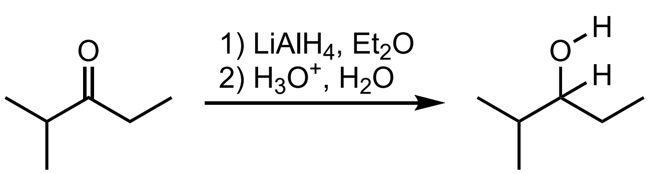


ketones yield secondary alcohols
Note: ternary mechanism not realistic

13 Aluminum Hydride Reduction of Ketones

mechanism ✓

8-5 pg.318

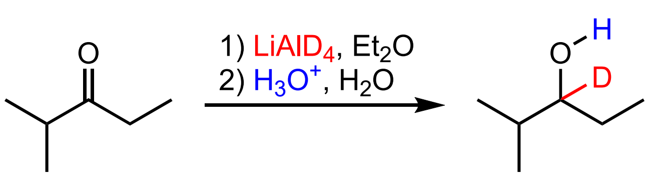


ketones yield secondary alcohols

14 Aluminum Deuteride Reduction of Ketones

mechanism ✓

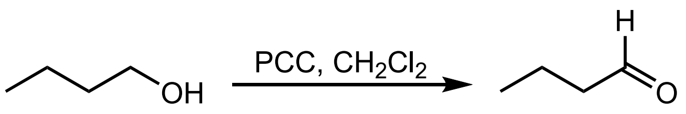
8-5 pg.318



ketones yield secondary alcohols
can label molecules with D
D added to carbonyl carbon $^{\delta+}C{=}O^{\delta-}$

15 PRIMARY ALCOHOLS WITH PCC

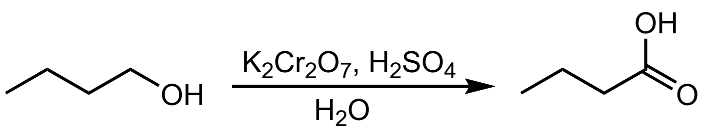
8-5 p 320



anhydrous conditions prevent formation of hydrate and further oxidation

16 PRIMARY ALCOHOLS WITH CHROMIC ACID

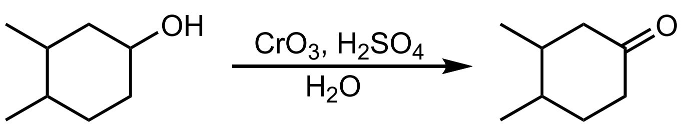
8-5 p 320



after oxidation to the aldehyde, hydrate forms in aqueous acidic conditions
hydrate undergoes further oxidation

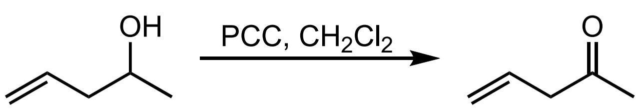
17 SECONDARY ALCOHOLS WITH CHROMIC ACID

8-5 p 320



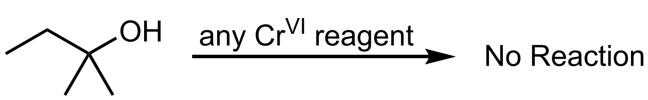
18 SECONDARY ALCOHOLS WITH PCC

8-5 p 320



19 TERTIARY ALCOHOLS

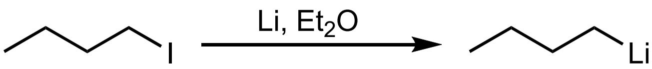
8-5



oxidation not possible

20 Synthesis of Alkyllithium Reagents

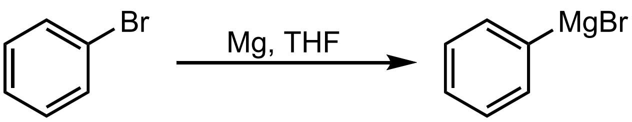
8-6



strongly polarized $^{\delta-}$C-Li$^{\delta+}$ bond

21 Synthesis of Grignard Reagents

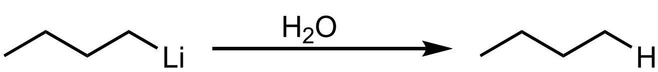
8-6



strongly polarized $^{\delta-}$C-Mg$^{\delta+}$ bond

22 Hydrolysis of Alkylithiums

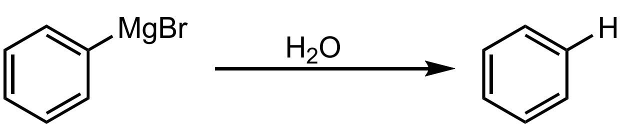
8-6



alkyllithiums are strongly basic
any protic species can neutralize (ROH, RCO_2H, RNH_2)

23 Hydrolysis of Grignards

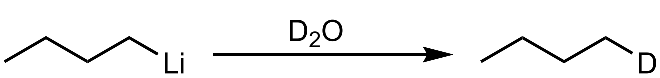
8-6



Grignards are strongly basic
any protic species can neutralize (ROH, RCO_2H, RNH_2)

24 Hydrolysis of Alkylithiums with D_2O

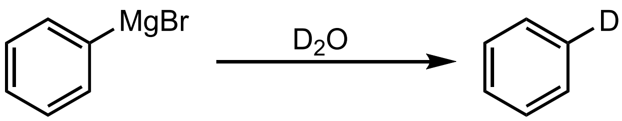
8-6



useful for labeling a molecule

25 Hydrolysis of Grignard with D_2O

8-6

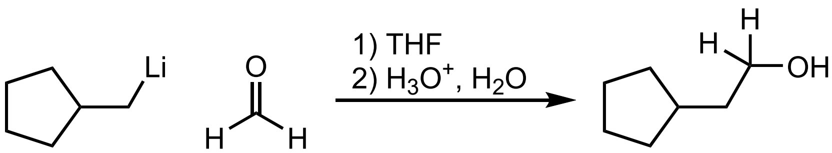


useful for labeling a molecule

26 RLi Addition to Formaldehyde

mechanism ✔
pg.325

8-7

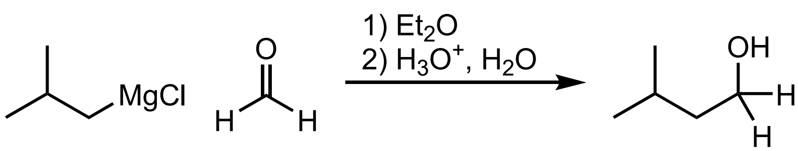


primary alcohol produced

27 Grignard Addition to Formaldehyde

mechanism ✔
pg.325

8-7

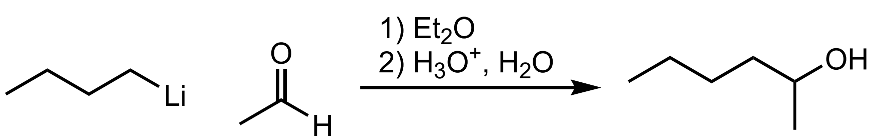


primary alcohol produced

28 RLi Addition to Aldehyde

mechanism ✔
pg.325

8-7

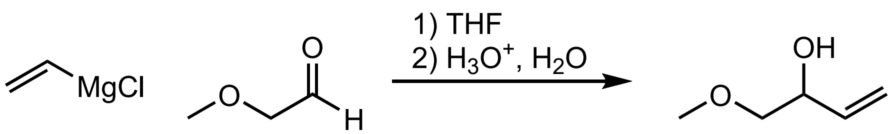


secondary alcohol produced
lookout for other reactive functional groups *e.g.* O–H or C=O

29 Grignard Addition to Aldehyde

mechanism ✓

8-7 pg.325

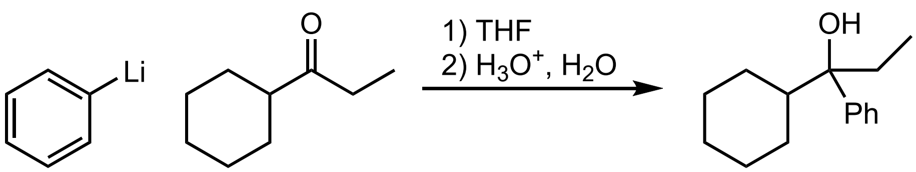


secondary alcohol produced
lookout for other reactive functional groups *e.g.* O−H or C=O

30 RLi Addition to Ketone

mechanism ✓

8-7 pg.325

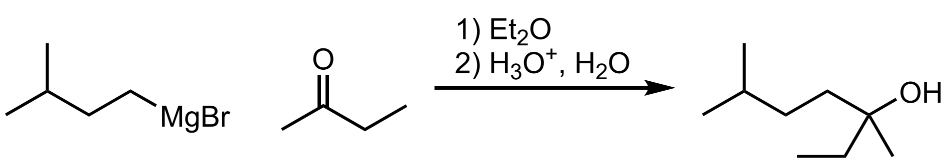


tertiary alcohol produced
lookout for other reactive functional groups *e.g.* O−H or C=O

31 Grignard Addition to Ketone

mechanism ✓

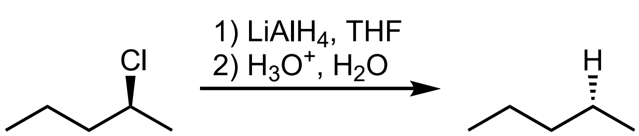
8-7 pg.325



tertiary alcohol produced
lookout for other reactive functional groups *e.g.* O−H or C=O

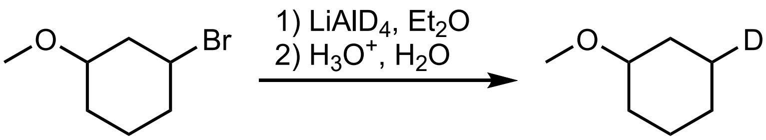
32 $LiAlH_4$ Addition to Haloalkane

8-6



33 LiAlD$_4$ Addition to Haloalkane

8-6



useful for labeling a molecule

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34 Deprotonation of Alcohols

8-3, 9-1

strong bases: LDA, *n*-BuLi, NaH, *etc.*
hydroxide (HO$^-$) only forms balanced equilibrium with alkoxide (RO$^-$)

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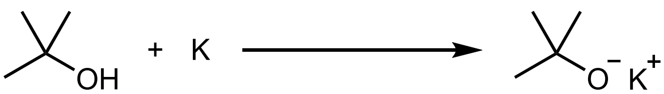
35 Alkali Metals with Alcohols

9-1



useful with Li, Na, and K
alcohol reactivity – tertiary < secondary < primary < methyl

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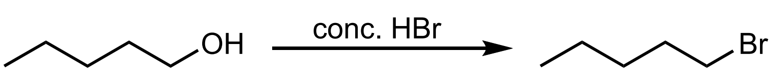
36 HX with Primary Alcohols

mechanism
pg.354

8-3, 9-2, 9-3



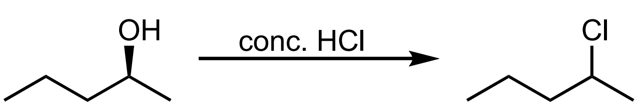
works with HBr or HI via S$_N$2
nucleophilic acids

---

37 HX with Secondary Alcohols

mechanism
pg.356

8-3, 9-2, 9-4



works with HCl, HBr, or HI via S$_N$1
nucleophilic acids
lookout for carbocation rearrangements

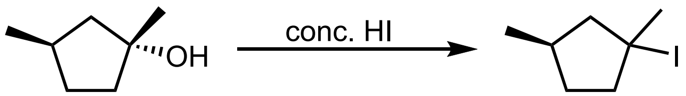
---

38 HX with Tertiary Alcohols

mechanism ✓

8-3, 9-2, 9-5

pg.356



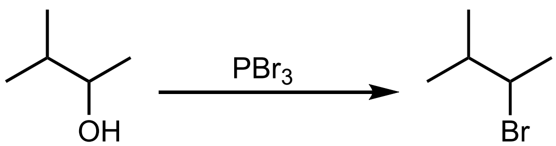
works with HCl, HBr, or HI via S_N1
nucleophilic acids
lookout for carbocation rearrangements

39 Bromoalkane Synthesis

mechanism ✓

9-4

pg.365



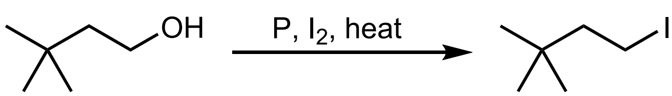
forms a phospite ester intermediate (good LG!)
works well with primary and secondary substrates via S_N2
no carbocation so no rearrangement

40 Iodoalkane Synthesis

mechanism ✓

9-4

pg.365

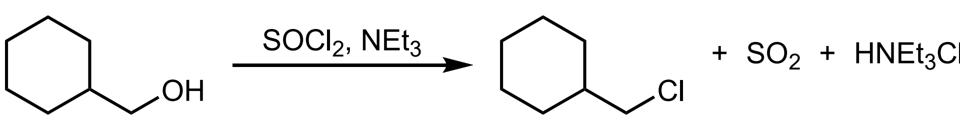


forms a phospite ester intermediate (good LG!)
works well with primary and secondary substrates via S_N2
no carbocation so no rearrangement

41 Chloroalkane Synthesis

9-4

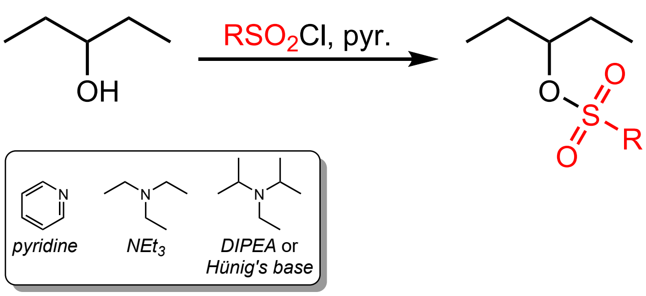
p
366

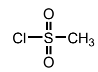
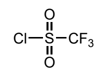
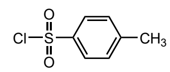


forms an inorganic ester intermediate (good LG!)
works well with primary and secondary substrates via S_N2
mild, non-nucleophilic base added to remove HCl such as pyridine, triethylamine (NEt_3), or diisopropylethylamine (DIPEA) aka Hünig's Base (see Sulfonate Synthesis 42 for examples)

42 Sulfonate Synthesis

9-4



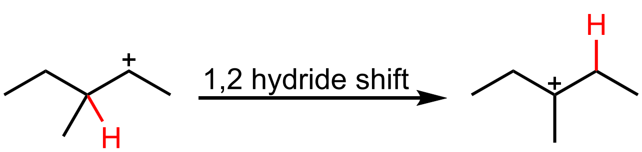
Structure	Abbv.	Fragment	RSO_3^- name
	MsCl	mesyl	mesylate
	TfCl	triflyl	triflate
	TsCl	tosyl	tosylate

create excellent leaving groups from alcohols

43 Hydride Shift

mechanism ✔

9-3 pg.358

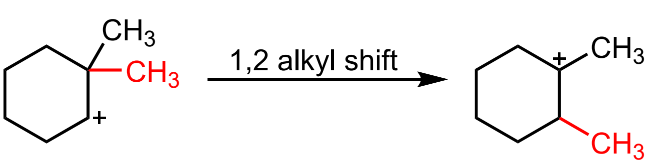


carbocation C^+ can form on secondary or tertiary positions
will move to tertiary if present and nearby
small E_A means a mixture of C^+ will be produced

44 Alkyl Shift

mechanism ✔

9-3 pg.361

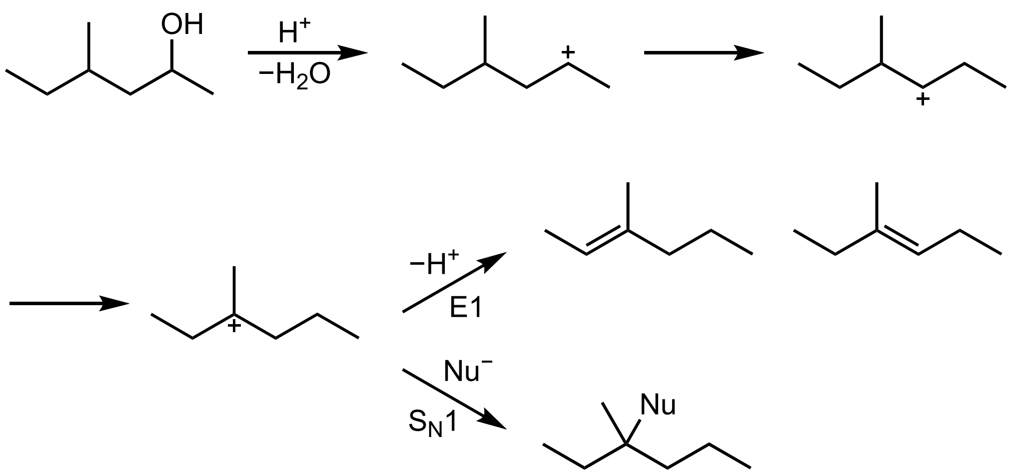


carbocation C^+ can form on secondary or tertiary positions
will move to tertiary if present and nearby
small E_A means a mixture of C^+ will be produced

45 Carbocation Rearrangement Reactions

9-3

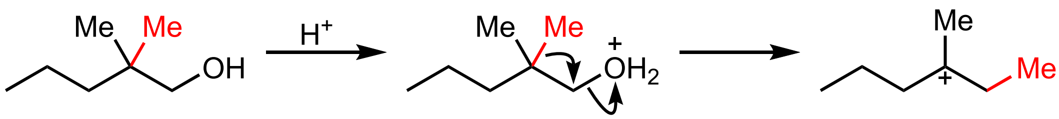
mechanism ✔
pg.358, 361



46 Concerted Alkyl Shift (Primary Leaving Groups)

9-3

mechanism ✔
pg.363

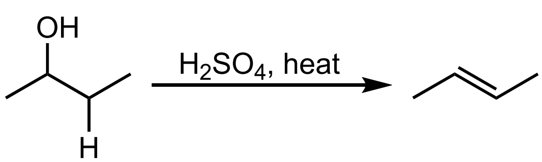


primary carbocations are too unstable to produce
adjacent quaternary carbons will form a tertiary carbocation through a concerted mechanism

47 Dehydration of Alcohols

9-2, 9-3, 9-7, 11-5

mechanism ✔
pg.356, 283, 285



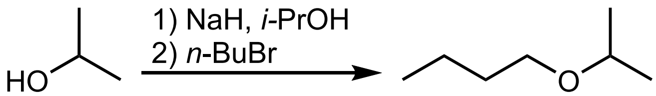
carbocation C^+ rearrangements can occur
as C^+ stability decreases, higher T required
primary ROH: 170–180°C, via E2 mechanism
secondary ROH: 100–140°C, usually via E1 mechanism
tertiary ROH: 25–80°C, via E1 mechanism

48 Williamson Synthesis of Primary Ethers

mechanism ✔

9-6

pg.373, 374



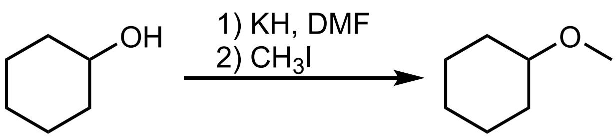
alkyl halide must be methyl or primary (often CH_3I)
alcohol can be primary or secondary; tertiary alkoxides give E2 products
intramolecular depends on ring size and is a balance of kinetic (smaller better) and thermodynamic (less ring strain better) factors
when forming cyclic ethers – $k_3 \geq k_5 > k_6 > k_4/geqk_7 > k_8$
k_n = reaction rate and n = ring size

49 Williamson Synthesis of Secondary Ethers

mechanism ✔

9-6

pg.373, 374



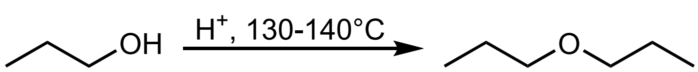
see notes in primary ether synthesis 48

50 Primary Ether Synthesis

mechanism ✔

9-7

pg.378



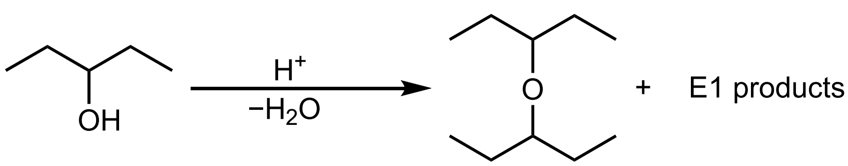
must use a non-nucleophilic acid
protonate one alcohol, S_N2 with another

51 Secondary Ether Synthesis

mechanism ✔

9-7

pg.379

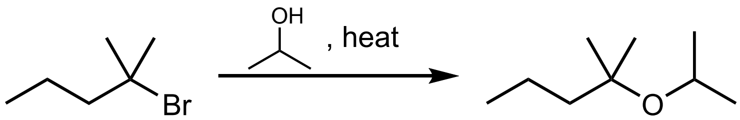


must use a non-nucleophilic acid
protonate one alcohol, S_N1 with another
competes with elimination

52 Tertiary Ether Synthesis by Solvolysis

9-7

mechanism ✓
pg.379



solvolysis – see hydrolysis (S_N1) 9
tertiary alcohols can be used with acid to form carbocation

53 Primary Ether Acidic Cleavage

9-8, 7-9

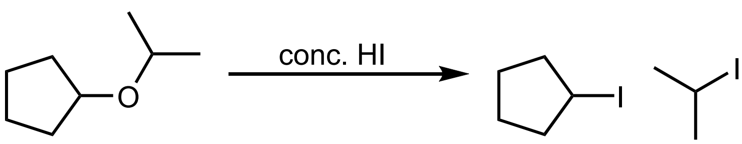
mechanism ✓
pg.380

can use HBr or HI
proceeds via S_N2 1 following protonation of the alcohol 5

54 Secondary Ether Acidic Cleavage

9-8, 7-9

mechanism ✓
pg.380, 271, 273

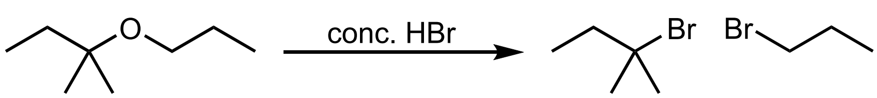


can use HBr or HI
proceeds via S_N2 or S_N1
protonation 5 followed by substitution 1 at primary carbon or 2 if both R groups are secondary or tertiary

55 Tertiary Ether Acidic Cleavage

9-8, 7-9

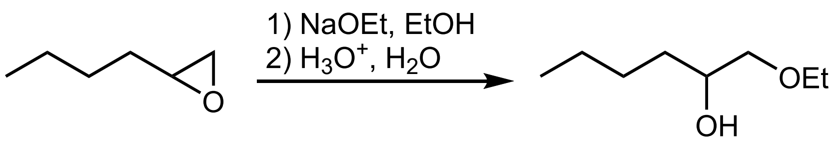
mechanism ✓
pg.380, 271, 273



can use HBr or HI
proceeds via S_N2 or S_N1
protonation 5 followed by substitution 1 at primary carbon or 2 if both R groups are secondary or tertiary

56 Epoxide Opening with Anionic Nucleophiles

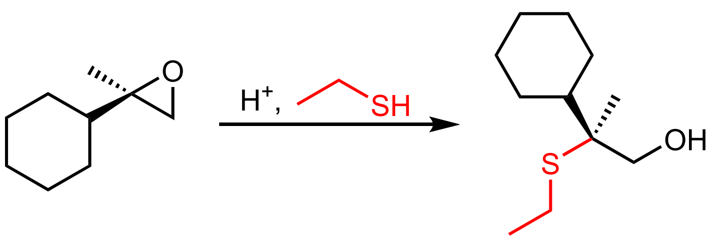
9-9 mechanism ✔ pg.383, 384



strong nucleophiles will attack the less hindered carbon

57 Acid-Catalyzed Epoxide Opening

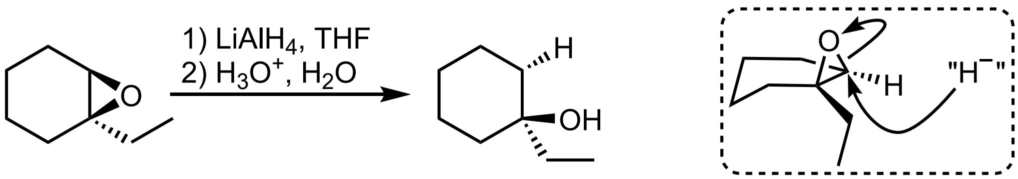
9-9 mechanism ✔ pg.385, 387



weak nucleophiles will attack the more substituted carbon
protonation 5 of the epoxide leads to δ^+ charges on each C of the ring
the more substituted carbon has a greater δ^+
stereospecific with chiral epoxides

58 Epoxide Opening with $LiAlH_4$

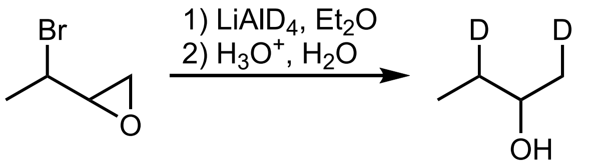
9-9



hydride addition to the less hindered carbon
stereospecific

59 Epoxide Opening with $LiAlD_4$

9-9



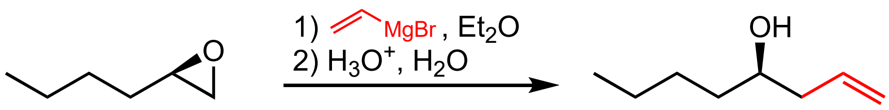
useful for labeling a molecule
D^- adds to the less hindered carbon
stereospecific

60 Epoxide Opening with Alkyllithium or Grignard Reagents

9-9

mechanism ✔

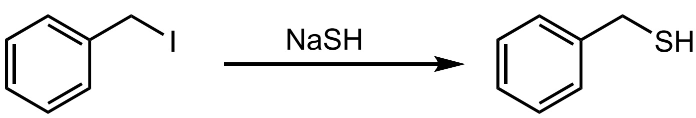
pg.383, 384



also with alkyllithium reagents (R–Li)
strongly nucleophilic so addition occurs at less hindered carbon
stereospecific with chiral epoxides

61 Thiol Synthesis

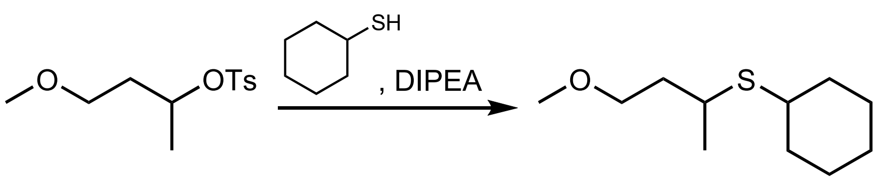
9-10



yet another appearance of an S_N2 reaction 1

62 Thioether Synthesis

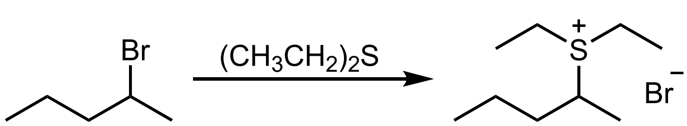
9-10



another S_N2 reaction 1 followed by deprotonation (reverse of 5)

63 Sulfonium Salt Synthesis

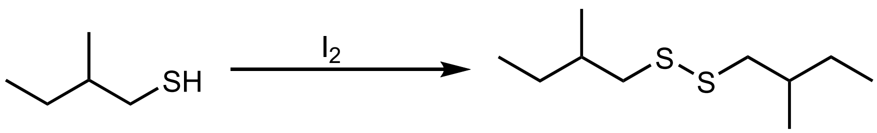
9-10



S_N2 reaction 1 with a dialkyl sulfide nucleophile
product is a sulfonium salt

64 Disulfide Synthesis by Thiol Oxidation

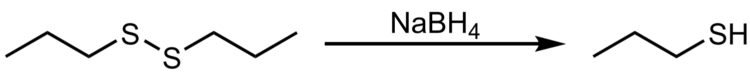
9-10



reversed with $NaBH_4$ in 65

65 Thiol Synthesis by Disulfide Reduction

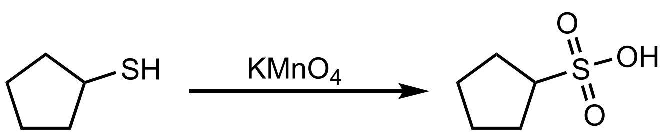
9-10



reversed with I_2 in 64

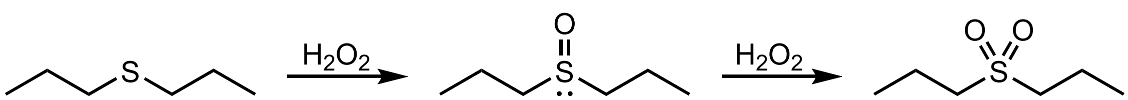
66 Sulfonic Acid Synthesis by Thiol Oxidation

9-10



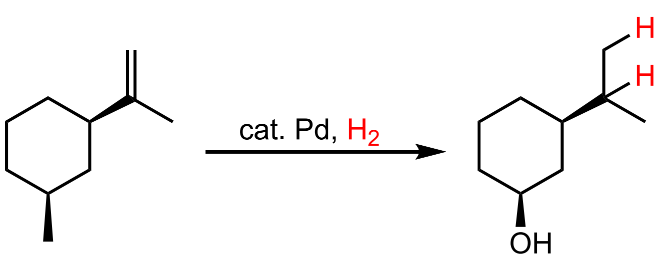
67 Thioether Oxidation

9-10



68 Hydrogenation of Alkenes

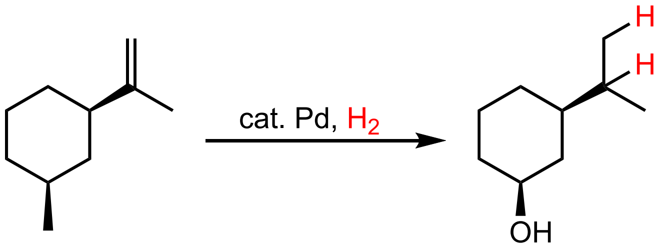
11-5



need a metal catalyst (Pd or Pt common, often as Pd–C or PtO_2)
exothermic
more substituted, more stable

69 Hydrogenation of Alkenes — Stereospecificity

11-5, 12-2



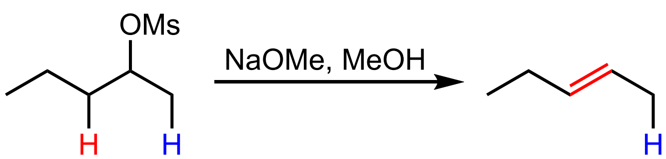
hydrogen atoms add to the same face of the alkene (*syn* addition)

70 E2 — Saytzev Elimination from an Unhindered Base

mechanism ✔

11-6

pg.490



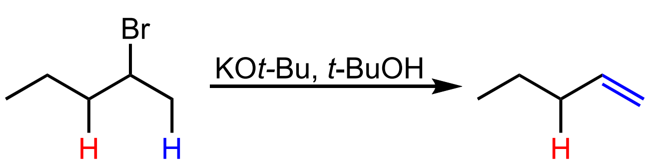
the more substituted, more stable alkene is formed
lower E_A associated with the more thermodynamically stable alkene

71 E2 — Hofmann Elimination from a Hindered Base

mechanism ✔

11-6

pg.490



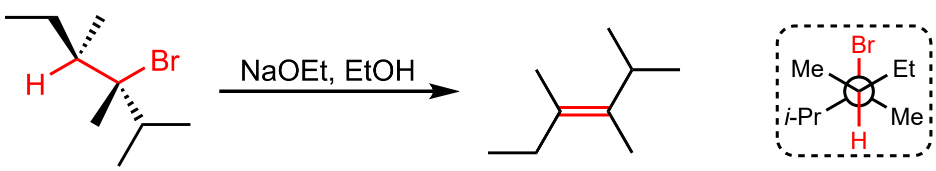
the less substituted, less stable alkene is formed
lower E_A associated with deprotonation of less substituted carbon due to lower steric interactions in the transition state

72 E2 — Stereochemistry and *anti* conformation

mechanism ✔

11-6

pg.493



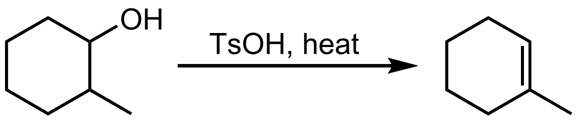
elimination occurs in *anti* configuration
stereospecific with diastereomers that have stereocenters at both reacting carbon atoms

73 Dehydration of Alcohols

mechanism ✔

11-7, 9-2, 9-7

pg.356



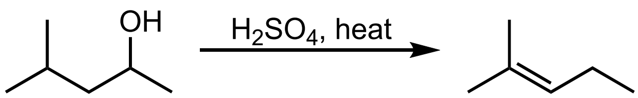
the more substituted, more stable alkene is formed

74 Carbocation Rearrangement during Dehydration

mechanism ✔

11-7, 9-3

pg.356, 358, 361



look out for rearrangements
the more substituted, more stable alkene is formed